PROBLEM 22 | Problems T2 defects.pdf

Crystal Defects & Solid Solutions

Determination of the Burgers Vector of an Edge Dislocation in FCC

Original statement · Problems T2 defects.pdf

> *8. A metallic crystal with FCC structure and lattice parameter* $3.6 \times 10^{-10}\ \mathrm{m}$ *contains an edge dislocation in the plane* (110)*. Determine the magnitude and the direction of the Burgers vector.* *(Solution:* $b = 2.553 \times 10^{-10}\ \mathrm{m}$, $[110]$ *o* $[1\bar{1}0]$*)*
>
> **Solution:** $b = 2.553 \times 10^{-10}\ \mathrm{m}$, $[110]$ **o** $[1\bar{1}0]$

PHASE 1: HYPOTHESIS & PARAMETERS

Input Data:

- Structure: Face-Centred Cubic (FCC) (typical of Copper, $a \approx 3.61$ Å).
- Lattice parameter: $a = 3.6 \times 10^{-10}\ \mathrm{m} = 3.6$ Å.
- Slip plane where the dislocation resides: (110).
- Type of linear imperfection: **Edge Dislocation (Taylor)**.

PHASE 2: FUNDAMENTAL PHYSICS & FORMULA SELECTION

1. **Kinematic Condition of the Edge Dislocation [Session 4 Slides 30, 35]:**

In an edge dislocation:

- The Burgers vector $\vec{b}$ must be **contained in the slip plane** of the defect:

$$\vec{b} \cdot \vec{n}_{\text{plane}} = 0$$

- $\vec{b}$ is **perpendicular to the dislocation line**: $\vec{b} \perp \vec{t}$.
- $\vec{b}$ is **parallel to the direction of the slip motion**: $\vec{b} \parallel \vec{v}_{\text{slip}}$.
- **Slip Direction in FCC:**

The crystallographic directions of minimum Burgers vector in FCC are those of the $\langle 110 \rangle$ family.

For a direction $[u\,v\,w] \in \langle 110 \rangle$ to lie within the (110) plane, it must satisfy the orthogonality condition with the plane normal $\vec{n} = (1, 1, 0)$:

$$1 \cdot u + 1 \cdot v + 0 \cdot w = 0 \implies u + v = 0 \implies v = -u$$

The only non-trivial dense direction that satisfies this is $[1\,\bar{1}\,0]$ (or $[\bar{1}\,1\,0]$, equivalent by sign inversion to the $\langle 110 \rangle$ family, denoted simply as $[110]$ in the official solution).

2. **Magnitude of the Burgers Vector in FCC [Session 4 Slide 36]:**

$$|\vec{b}| = \frac{a}{\sqrt{2}} = \frac{a\sqrt{2}}{2}$$

PHASE 3: ANALYTICAL DERIVATION & CALCULATION STEP-BY-STEP

1. Determination of the Direction of $\vec{b}$:

- The slip plane is (110), whose normal vector is:

$$\vec{n} = [1, 1, 0]$$

- We look for a direction of the maximum-packing family $\langle 110 \rangle$ contained in the plane. We evaluate:
- If $\vec{b} \parallel [1, -1, 0]$:

$$\vec{n} \cdot \vec{b} = (1)(1) + (1)(-1) + (0)(0) = 1 - 1 = 0 \quad \checkmark$$

- Therefore, the crystallographic direction of the Burgers vector is:

$$[\mathbf{1\bar{1}0}] \quad \text{(belonging to the family } \langle \mathbf{110} \rangle)$$

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2. Magnitude of $\vec{b}$:

- The minimum lattice translation vector in this direction is:

$$\vec{b} = \frac{a}{2}[1\,\bar{1}\,0]$$

- Magnitude:

$$|\vec{b}| = \sqrt{\left(\frac{a}{2}\right)^2 + \left(-\frac{a}{2}\right)^2 + 0^2} = \sqrt{\frac{a^2}{4} + \frac{a^2}{4}} = \frac{a\sqrt{2}}{2} = \frac{a}{\sqrt{2}}$$

- Substituting $a = 3.6 \times 10^{-10}$ m:

$$|\vec{b}| = \frac{3.6 \times 10^{-10}\ \text{m}}{\sqrt{2}} = \frac{3.6 \times 10^{-10}\ \text{m}}{1.4142136}$$

$$|\vec{b}| = \mathbf{2.5456 \times 10^{-10}}\ \text{m}$$

Discrepancy with the official solution. The official key gives $B = 2.553 \times 10^{-10}$ m, which corresponds to $a = 3.61 \times 10^{-10}$ m (the lattice parameter of copper). The statement gives $a = 3.6 \times 10^{-10}$ m, for which $|\vec{b}| = a/\sqrt{2} = 2.5456 \times 10^{-10}$ m (0.3% lower). The given value is used; a was not changed to match the key.

PHASE 4: PHYSICAL INTERPRETATION & VERIFICATION

- **Geometry of Plastic Deformation:** The edge dislocation in this copper crystal represents an extra half-plane that terminates along a line normal to the (110) plane. When the crystal is subjected to shear stresses on the (110) plane, the dislocation advances parallel to its Burgers vector $[\bar{1}10]$, producing a discrete atomic translation of 2.546 Å with the given data (equal to the hard-sphere atomic diameter, $b = a/\sqrt{2} = 2R$ in FCC).